

Plants in limbo – the theory of detrital photosynthesis

Journal:	<i>Quantitative Plant Biology</i>
Manuscript ID	QPB-2026-0013.R2
Manuscript Type:	Theories
Date Submitted by the Author:	n/a
Complete List of Authors:	Wright, Luka; UWA Oceans Institute, ; UWA School of Biological Sciences,
Keywords:	decay, degradation, leaf litter, blue carbon, biogeochemistry
Abstract:	Kelp detritus can maintain photosynthesis following detachment. This could be explained by limited tissue differentiation in non-vascular plants, but it remains unclear if detrital photosynthesis is restricted to macroalgae. Here I formalise the theory of detrital photosynthesis by developing mathematical and causal models and performing a meta-analysis of post-detachment physiology across 129 studies on 92 species of terrestrial and aquatic plants, including original data on the seagrasses <i>Amphibolis antarctica</i> and <i>Halophila ovalis</i>. Detrital photosynthesis is a general phenomenon, but its longevity in macroalgae is unmatched. Median photosynthetic half-life increases from terrestrial plants (11 days) over freshwater Alismatales (14 days) and seagrasses (17 days) to seaweeds (44 days) and only matches or exceeds detrital half-life for seaweeds (3:1). Detrital photosynthesis therefore uniquely determines macroalgal as opposed to vascular plant decomposition. This explains why light slows kelp decomposition, contrary to the expected effect of photodegradation. The presented theory questions conventional understanding of decomposition and the carbon cycle.



How long can detritus photosynthesise and how affected is its decomposition?

Terrestrial plants

Freshwater plants

Seagrasses

Seaweeds

Day Week Month Year Scarcely Partly Fully

Cambridge University Press



Quantitative Plant Biology

Theories

Plants in limbo — the theory of detrital photosynthesis

Luka Seamus Wright^{1,2*}

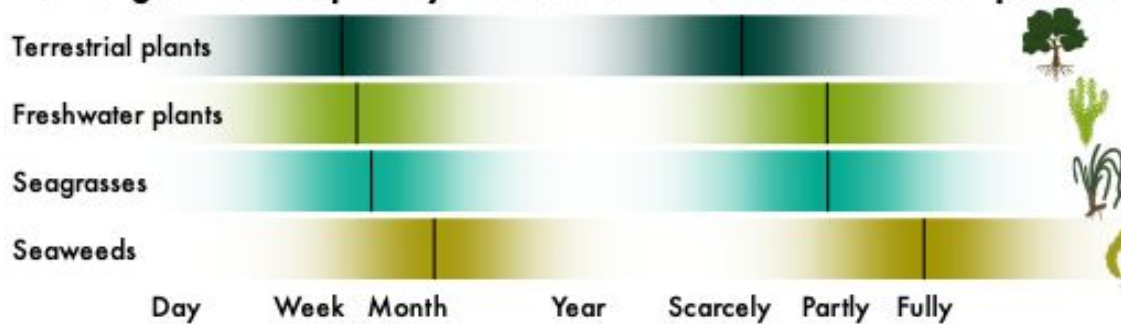
¹Oceans Institute, University of Western Australia, Perth, Australia

²School of Biological Sciences, University of Western Australia, Perth, Australia

*luka.wright@research.uwa.edu.au, luka@wright.it

Abstract

How long can detritus photosynthesise and how affected is its decomposition?



Kelp detritus can maintain photosynthesis following detachment. This could be explained by limited tissue differentiation in non-vascular plants, but it remains unclear if detrital photosynthesis is restricted to macroalgae. Here I formalise the theory of detrital photosynthesis by developing mathematical and causal models and performing a meta-analysis of post-detachment physiology across 129 studies on 92 species of terrestrial and aquatic plants, including original data on the seagrasses *Amphibolis antarctica* and *Halophila ovalis*. Detrital photosynthesis is a general phenomenon, but its longevity in macroalgae is unmatched. Median photosynthetic half-life increases from terrestrial plants (11 days) over freshwater Alismatales (14 days) and seagrasses (17 days) to seaweeds (44 days) and only matches or exceeds detrital half-life for seaweeds (3:1). Detrital photosynthesis therefore uniquely determines macroalgal as opposed to vascular plant decomposition. This explains why light slows kelp decomposition, contrary to the expected effect of photodegradation. The presented theory questions conventional understanding of decomposition and the carbon cycle.

Keywords decay, degradation, leaf litter, biogeochemistry, blue carbon

Introduction

...tossed and swept like dead leaves from one spot to another, never resting, never giving back their goodness to the earth, never fully dead but in some vast limbo between life and extinction, tossing and tumbling without end... — Philip Pullman (1994) *The Tin Princess*

Seaweed detrital dynamics are influenced by detrital photosynthesis. Detrital photosynthesis is defined as the persistence of photosynthesis after detachment (Frontier et al. 2021, Wright et al. 2022, 2024). Kelp detritus can remain photosynthetic for months (de Bettignies et al. 2020, Frontier et al. 2021, Wright & Foggo 2021, Frontier et al. 2022, Wright et al. 2022, Wright & Kregting 2023, Wright et al. 2024) and seaweed rafts sink before they stop photosynthesising (Graiff et al. 2013, 2016, Tala et al. 2019, Rothäusler et al. 2020). This results in atypical decomposition trajectories with a distinct lag in the onset of decay (Birch et al. 1983, Bourguès et al. 1996, Brouwer 1996, Vandendriessche et al. 2007, Frontier et al. 2021, 2022, Kennedy & Blain 2025). The fundamental effect on detrital dynamics becomes apparent when comparing live with dead or senescent seaweed detritus which decomposes up to an order of magnitude faster (Birch et al. 1983, Brouwer 1996, Hamersley et al. 2015, de Bettignies et al. 2020, White & Norkko 2025). This evidence stands in stark contrast to the established ecological definition of detritus as “dead organic matter” (Moore et al. 2004, discussed in Wright & Kregting 2023). One solution to this conundrum is to insist that detached plant material is not detritus as long as it is still alive. However, detached plants (except for holopelagic and pleustophytic species) are commonly referred to as detritus. For most plants, detachment provides a clear demarcation of the detrital phase which the ensuing photosynthetic limbo does not. By way of compromise, I continue to use the term “detritus” for any detached plant tissue and “detritus sensu stricto” for detached *and* dead plant tissue.

Detrital photosynthesis underpins other aspects of physiology. Remarkably, seaweed may continue to grow after detachment (Norton & Mathieson 1983, Vandendriessche et al. 2007, Rothäusler et al. 2011, Tala et al. 2016, Rothäusler et al. 2018, Frontier et al. 2021, Pedersen et al. 2021, Filbee-Dexter et al. 2022, Frontier et al. 2022, Wright et al. 2022). This suggests that detrital photosynthesis can counteract decomposition by supporting tissue growth and repair. Moreover, chemical defences are maintained after detachment. Excised seaweed tissues are

capable of oxidative bursts (Weinberger et al. 1999, Weinberger & Friedlander 2000, Küpper et al. 2001, 2002, Thomas et al. 2014) and constitutive hydrogen peroxide release (Küpper et al. 2002, Twigg et al. 2020), which inhibit heterotrophic bacteria (Weinberger & Friedlander 2000, Küpper et al. 2001, 2002, Twigg et al. 2020) and grazers (Thomas et al. 2014). Excised brown algal tissues additionally produce phlorotannins which aid wound healing besides acting as chemical defence (Lüder & Clayton 2004, Koivikko et al. 2005, Halm et al. 2011). Steady and increasing phlorotannin content of kelp detritus (Tala et al. 2013, 2016, 2019, de Bettignies et al. 2020, Wright et al. 2022, but see Pedersen et al. 2021) may thus be interpreted as a response to decomposition (Wright et al. 2022). This suggests that detrital photosynthesis can also counteract decomposition by enabling chemical defence. Finally, there is overwhelming evidence for reproductive kelp detritus (Macaya et al. 2005, Hernández-Carmona et al. 2006, Rothäusler et al. 2011, Tala et al. 2013, Fraser et al. 2018) and persistence or development of reproductive structures in experimentally detached seaweeds (Hernández-Carmona et al. 2006, Rothäusler et al. 2011, Tala et al. 2016, Rothäusler et al. 2018, Tala et al. 2019, de Bettignies et al. 2020, Rothäusler et al. 2020, Frontier et al. 2021). Detritus can therefore act as a propagule, often better suited to long-distance dispersal than spores and gametes (Norton 1992, Fraser et al. 2018). Even seaweed detritus that is not reproductive can be a means of dispersal through vegetative propagation (Norton & Mathieson 1983), which is also common among seagrasses (Hall et al. 2006, Stafford-Bell et al. 2015, Weatherall et al. 2025, Hanuise et al. 2026). Detrital photosynthesis, providing the energy allocated to growth, repair, defence and reproduction, must thus be regarded as evidence of detrital physiology more broadly.

There is currently no reference point for detrital photosynthesis. The term was first introduced quite recently in the context of kelp decomposition (Frontier et al. 2021, Wright et al. 2022, 2024). But there was clearly awareness of the phenomenon among early phycologists who noted the existence of “living litter” (Norton & Mathieson 1983) and ability of excised tissue to maintain photosynthesis (Drew 1983), and deemed it necessary to “pre-kill” detritus to measure decomposition (Josselyn & Mathieson 1980, Rice & Tenore 1981, Birch et al. 1983, Smith & Foreman 1984, Brouwer 1996). The term “detrital photosynthesis” is not used in other fields of botany but the amount of research into photosynthesis of excised leaves of terrestrial and freshwater plants is enormous (e.g. Thimann & Satler 1979, Jana & Choudhuri 1980, 1982, Kar

& Choudhuri 1986, 1987, Powles et al. 2006, Gauthier & Jacobs 2018, Kar et al. 2021). The literature also includes three studies on seagrasses (Knauer & Ayers 1977, Pellikaan 1982, Strother & Vatta 1986), the only fully marine angiosperms and vascular counterpart to seaweeds (Olsen et al. 2016). The aims of measuring photosynthesis in detached leaves are varied, with most studies using excision as an experimental proxy for plant senescence. No attempt has been made to align these seemingly incongruous data in a formal comparison across plant groups (sensu Bolton 2016). It consequently remains unclear if detrital photosynthesis is unique to seaweeds. Although it has been suggested that detrital photosynthesis may be explained by the ability of all macroalgal tissues to independently photosynthesise to some extent (Wright & Kregting 2023), there is no certainty due to lack of comparison with other plants. The mechanism underlying the ability of seaweeds to maintain detrital photosynthesis remains unclear. It could simply be their aquatic nature which removes reliance on roots for water and nutrient provisioning. Or it could be more deeply rooted in their non-vascularity which requires minimal tissue differentiation. To understand the prerequisites for detrital photosynthesis and thus predict prevalence among plants, we need to look beyond seaweeds.

Seagrasses are suitable candidates for further investigation of detrital photosynthesis. While they are also marine macrophytes, seagrasses are fundamentally different from seaweeds. Seagrasses are vascular plants which, due to their terrestrial ancestry (Olsen et al. 2016), are differentiated into roots, rhizomes or stems, and leaves. However, convergent evolution with seaweeds (Olsen et al. 2016) has reduced their reliance on roots for nutrient uptake. Seagrass leaves tend to outperform their roots in supplying nutrients to the plant (Pedersen & Borum 1992, Stapel et al. 1996, Pedersen et al. 1997, Gras et al. 2003, Viana et al. 2019). This suggests that seagrass leaves can independently photosynthesise, seemingly reducing the primary function of the roots to anchorage, analogous to the holdfast of seaweeds. However, roots can supply more nitrogen despite the leaves' greater uptake affinity because of the higher ammonium concentration in sediment pore water (Short & McRoy 1984, Lee & Dunton 1999, but see Pedersen & Borum 1992) and tap into nutrient pools that are unavailable to leaves, such as particulate (Evrard et al. 2005) and microbially fixed (Patriquin & Knowles 1972, Mohr et al. 2021) nitrogen. Nutrients are also translocated from roots to leaves (Short & McRoy 1984, Viana et al. 2019), even those tens of centimetres from the source (Marbà et al. 2002). Clearly the photosynthetic independence

of seagrass leaves is at a halfway point between terrestrial plants and macroalgae, so seagrasses probably display detrital photosynthesis to some extent. Relatively slow and light-dependent chlorophyll degradation in seagrass detritus (Knauer & Ayers 1977, Pellikaan 1982, Strother & Vatta 1986) provides some support this. Seagrass detritus also often includes green leaves and vegetative fragments (Knauer & Ayers 1977, Ochieng & Erftemeijer 1999, Stafford-Bell et al. 2015, Jiménez-Ramos et al. 2023, Weatherall et al. 2025), likely owing to premature detachment (Cebrián et al. 1997, Hall et al. 2006, Jiménez-Ramos et al. 2023, Weatherall et al. 2025) and low nutrient resorption prior to abscission (Harrison 1989, Pedersen & Borum 1992, Pedersen et al. 1997, Hemminga et al. 1999, Rosch & Koch 2009). The fragments are typically viable and may remain so for a few weeks (Hall et al. 2006, Stafford-Bell et al. 2015, Weatherall et al. 2025, Hanuise et al. 2026). However, seagrass detrital photosynthesis remains to be directly investigated.

Here I aim to formalise the theory of detrital photosynthesis. I hope that this will provide a better conceptual foundation for the investigation of plant decomposition and carbon cycling more generally. To address this aim, I approached detrital photosynthesis holistically for the first time. I conducted a brief ex situ decomposition experiment with seagrass leaves to fill the knowledge gap of seagrass detrital photosynthesis. I chose the paddle weed *Halophila ovalis* (Hydrocharitaceae) and wire weed *Amphibolis antarctica* (Cymodoceaceae) as diverse representatives because of their phylogenetic (diverged 105 Ma ago, Waycott et al. 2018) and morphological (de los Santos et al. 2012, 2016) differences. Senescence of fresh leaves was induced by excision and leaving them to decompose on sediment, measuring net oxygen production of destructively sampled leaves at regular intervals over 34 days (see *Supplementary methods*). On the basis of these and previous observations (Frontier et al. 2021, 2022, Wright et al. 2022, 2024), I then developed a mathematical model which describes detrital photosynthesis as a logistic decay function of detrital age. After compiling a detrital photosynthesis dataset (Wright 2025), I used this model in a Bayesian meta-analysis to infer the longevity of detrital photosynthesis for different plants and quantify the effects of light and choice of response variable (photosynthesis vs. chlorophyll). I compared the persistence of detrital photosynthesis with detrital half-life (sensu Olson 1963) to estimate the relative influence of physiology on

decomposition for different plants. Finally, I present a new causal model of decomposition that emerges naturally from the theory of detrital photosynthesis.

Methods

Model

Detached plant parts can remain in limbo for long durations, often photosynthesising at similar rates to the attached plant. For this reason and for simplicity, detrital photosynthesis has mostly been described as a linear function of detrital age (Frontier et al. 2021, Wright & Foggo 2021, Wright et al. 2022, Wright & Kregting 2023, Wright et al. 2024). But in two cases a logistic decay function was used: in a beta model predicting chlorophyll fluorescence (F_v/F_m) (Frontier et al. 2022) and in a binomial model predicting the probability of photosynthesis (Wright et al. 2024) with time after excision. The logistic function describes the decay of photosynthesis after detachment better than the linear function because it is bounded. Detrital photosynthesis cannot conceivably exceed baseline photosynthesis under identical conditions. An ill-defined baseline or changes in environment (e.g. when detritus floats and is exposed to more light) may naturally lead to brief increases after detachment (Frontier et al. 2021, Wright & Kregting 2023). However, this cannot be considered part of the photosynthetic decay process to be modelled. Similarly, photosynthesis cannot decline beyond a set minimum, which is zero for most response variables (e.g. gross photosynthesis, chlorophyll fluorescence and chlorophyll) and a negative constant for net photosynthesis. The logistic function is therefore the natural choice.

There are various ways to parameterise the logistic function. Typically, a simple linear term with intercept and slope is inserted into the standard logistic function. But for logistic decay with time (t), the intercept is not a meaningful parameter because $f(t=0)$ is the known baseline. The slope is more meaningful because it provides the logistic rate of decay per unit t , but ultimately the parameter of interest is the sigmoid inflection point or midpoint: t when $f(t) = 0.5$. It has the same units as t and is the only parameter that gives a directly interpretable measure of decay. As such it is analogous to half-life ($t_{1/2}$), t when $f(t) = 0.5$ for the exponential decay function (Olson 1963). The logistic midpoint can be calculated as intercept divided by slope, but this usually results in enormous uncertainty and difficulty to calculate meaningful central tendencies (Wright et al. 2024), so the midpoint is best directly estimated. The slope and midpoint parameterisation is

sometimes used but these parameters are not identifiable with an intercept near a known baseline. To simplify the model while retaining the midpoint, one must either fix the slope or intercept. Only a constant intercept is sensible and since the baseline is expected near 1, a logistic intercept of 5, translating to $f(0) = 0.99$, is the natural choice.

The most complex case to model is that of net photosynthesis since it has a negative lower bound. Two scaling parameters outside the standard logistic function are required. Thus my full logistic function describing the decay of photosynthesis (p) with time after detachment (t) is

$$p(t) = \frac{\alpha + \tau}{1 + e^{\frac{5(t-\mu)}{\mu}}} - \tau \quad (1)$$

where α is baseline photosynthesis, μ is the midpoint of photosynthetic decay and τ is baseline respiration. $\tau = 0$ when the lower bound is zero (e.g. gross photosynthesis) and $\alpha = 1$ when photosynthesis is normalised relative to the baseline. The naming of parameters is inspired by the word חַיָּה (graecised $\tau\mu\alpha$) which is composed of the first, middle and last letters of the Hebrew alphabet, symbolising life (ח), transition (ט) and death (א). In the story of the golem (McElreath 2020), the word lends life but removing ח changes its meaning from truth to death (תא) and takes life.

Literature review

Following the seagrass experiment (see *Supplementary methods*), I searched the literature using keyword strings such as “(photosynthesis OR chlorophyll) AND (‘induced senescence’ OR detrit* OR detach* OR excis* OR cut OR isolate*) AND (week* OR day* OR hour* OR minute* OR time OR age)” in Web of Science (webofscience.com/wos) and Google Scholar (scholar.google.com). Once I had a selection of suitable papers, I used Crossref Metadata Search (search.crossref.org) to find similar papers. I accepted studies on all plants (sensu Bolton 2016), the condition being that senescence was induced by excision at an exact time and photosynthesis was repeatedly measured in a timeseries of at least 3 timepoints over any interval to assess viability of the detached tissue. Studies on abscission were not considered since senescence starts at an unknown time prior to detachment. I naturally did not include plants that always live detached, such as pleustophytes and the holopelagic seaweeds *Sargassum fluitans* and *S. natans*,

for which detrital age cannot be defined as time since detachment. I accepted various response variables, including O₂ and CO₂ gas exchange, ¹⁴C assimilation, Fv/Fm, photosystem I and II electron transport rates (Mehler and Hill reactions), RuBisCo activity or concentration, carbonic anhydrase activity and photorespiration (glycolate concentration), as well as total chlorophyll, chlorophyll *a* or a chlorophyll indicator. Microbial photosynthesis was generally assumed to be negligible since most of the mentioned response variables would likely not detect it. Including the present study, this resulted in 129 suitable studies between 1957 and 2025 (see *Supplementary references*) and 551 experiments (median = 3 per study).

Data were extracted by copying tables or digitising figures using WebPlotDigitizer v4.6 (Rohatgi 2022). In two cases I contacted authors for their raw data and in five cases the data were my own. Raw data were otherwise not available. The resulting dataset (Wright 2025) contains data on 92 species from 23 orders in four non-taxonomic groups of interest: terrestrial plants (66 species), freshwater plants (6 species), seagrasses (5 species) and seaweeds (15 species). It is noteworthy that all freshwater plants in the dataset belong to the order Alismatales, like seagrasses, and all seaweeds belonged to the phyla Chlorophyta and Heterokontophyta (Table S1). The assigned taxonomy follows Plants of the World Online (powo.science.kew.org, POWO 2025) and AlgaeBase (algaebase.org, Guiry, M.D. & Guiry 2025). My distinction between freshwater and terrestrial plants is based on whether leaves are submerged and photosynthesis can therefore be considered aquatic. There were no studies on freshwater macroalgae, so seaweeds are assumed to be representative. Several predictor variables including light- and water availability as well as the type of photosynthesis measurement were also recorded. When individual observations could not be extracted, the mean, standard error of the mean (s.e.m.) and sample size (n) were recorded instead. Uncertainty was mostly given as s.e.m., but when standard deviation (sd) was provided instead, it was converted as $\text{sem} = \frac{\text{sd}}{\sqrt{n}}$. In a few cases uncertainty was given as a 95% confidence interval (CI) or interquartile range (IQR). In the former case, I converted as $\text{sem} = \frac{0.5 \times \text{CI}_{0.95}}{\Phi^{-1}(0.975)}$, in the latter I assumed mean = median and conservatively converted the larger of the two quartile ranges (Q₃ – Q₂ or Q₂ – Q₁) as $\text{sem} =$

$$\frac{\text{QR}}{\Phi^{-1}(0.75) \times \sqrt{n}}$$

Meta-analysis

As is typical of meta-analyses, the data contain a mixture of observations and means. To jointly analyse data with at different levels of uncertainty one must either summarise observations and build a measurement error model or expand means to observations by simulation. I opted for the latter and obtained 10646 observations by simulating draws from the sampling distribution. For most response variables, which are bounded by zero, I chose the gamma distribution, for net photosynthesis I chose the normal distribution and for F_v/F_m I chose a truncated normal distribution because some summary statistics violated the beta distribution. I implemented the sampling in R v4.5.2 (R Core Team 2026) using “`rgamma( n , mean^2 / sd^2 , mean / sd^2 )`”, “`rnorm( n , mean , sd )`” and “`rtnorm( n , mean , sd , 0 , 1 )`”, the last function being from `extraDistr` v1.10.0 (Wolodzko 2023).

The response variable is an amalgam of various measures of photosynthesis. I therefore normalised it to enable formal comparison. Observations were expressed relative to the initial observation or mean of initial observations in the timeseries. The resulting ratios, representing relative photosynthesis, are the least invasive form of normalisation and frequently used to express photosynthetic decay (29 of 129 studies, e.g. Thimann & Satler 1979, Jana & Choudhuri 1980, Pellikaan 1982). The drawback of this approach is that data are not fully brought onto the same scale because net photosynthesis allows zeros and negative ratios. I resolved this issue by replacing all zeros and negative ratios with the minimum observed positive ratio, assumed to be the minimum observable ratio within measurement error of zero. This means equating net respiration with absence of photosynthesis, which is of course not the case because respiration can mask photosynthesis in net photosynthesis measurements. Nonetheless, I believe this to be acceptable for the purposes of determining the timescales of photosynthetic decay. This normalisation allows using the simplest form of Equation 1 with $\tau = 0$ and $\alpha = 1$.

Apart from t (days), the main predictor variables of interest are plant group (G = terrestrial plants, freshwater plants, seagrasses and seaweeds), light availability and the type of response variable (chlorophyll or photosynthesis). Light is naturally a key predictor of photosynthetic persistence (Frontier et al. 2021, 2022, Wright et al. 2024) and the type of response variable is important because chlorophyll and photosynthesis are expected to senesce on different

timescales. Of the recorded predictors, water availability is only relevant to terrestrial plants and was therefore not considered in the model. Data coverage is incomplete across the main predictor variables, with data missing for the combinations freshwater–light–photosynthesis, seagrass–dark–photosynthesis and seaweed–dark–chlorophyll. Since light availability and the type of response variable are binary predictors, they were coded as indicators of light ($L = -0.5$ for dark, $+0.5$ for light) and chlorophyll ($C = -0.5$ for photosynthesis, $+0.5$ for chlorophyll), centred on zero to avoid the bias in prior variance typical of indicator variables (McElreath 2020) and allow easy prediction of the mean. To get a sense of interspecific and experimental variation, I also included plant species (S , 92 levels) and experiment (E , 551 levels) as predictors. The resulting Bayesian log-normal multilevel model of relative photosynthesis (p) in standard statistical notation (sensu McElreath 2020) is

$$\ln p_i \sim \text{Normal}(\bar{p}_i, \sigma) \quad (2)$$

$$\bar{p}_i = \ln \frac{1}{1 + e^{\frac{5(t_i - \mu_i)}{\mu_i}}}$$

$$\ln \mu_i = \alpha_{G_i} + \alpha_{S_i} + \alpha_{E_i} + \beta_{G_i} L_i + \gamma_{G_i} C_i$$

$$\alpha_G \sim \text{Normal}(\bar{\alpha}, \sigma_{\alpha_G})$$

$$\alpha_S \sim \text{Normal}(0, \sigma_{\alpha_S})$$

$$\alpha_E \sim \text{Normal}(0, \sigma_{\alpha_E})$$

$$\beta_G \sim \text{Normal}(\bar{\beta}, \sigma_{\beta})$$

$$\gamma_G \sim \text{Normal}(\bar{\gamma}, \sigma_{\gamma})$$

$$\bar{\alpha} \sim \text{Normal}(\ln 14, 0.3)$$

$$\bar{\beta}, \bar{\gamma} \sim \text{Normal}(0, 0.4)$$

$$\sigma_{\alpha_G}, \sigma_{\alpha_S}, \sigma_{\alpha_E} \sim \text{Normal}_{0,\infty}(0, 0.3)$$

$$\sigma_{\beta}, \sigma_{\gamma} \sim \text{Normal}_{0,\infty}(0, 0.4)$$

$$\sigma \sim \text{Exponential}(1)$$

where the logarithm of μ (Equation 1) is described by a linear model with intercepts (α) indexed by G , S and E along with the effects of light (β) and measuring chlorophyll instead of photosynthesis (γ) indexed by G . β and γ are differences on the logarithmic scale, so represent log ratios. All parameters are partially pooled to enable prediction for new categories (e.g.

experiments, species, plant groups), borrow information and regularise spurious effects (McElreath 2020). The intercept standard deviations (σ_a) give an indication of intergroup, interspecific and experimental variance. The prior for the mean of μ on the logarithmic scale ($\bar{\alpha}$) is centred on $\ln 14$ because two weeks seems like a reasonable median midpoint of photosynthetic decay based on my observations on kelps (Wright et al. 2022, 2024) and literature review. Even though I suspected β and γ to be positive, I centred the priors for $\bar{\beta}$ and $\bar{\gamma}$ on zero to avoid biasing the effect.

The model was written in Stan (Carpenter et al. 2017) and implemented with CmdStan v2.17.1 (Lee et al. 2026) in R. This was facilitated by cmdstanr v0.9.0 (Gabry et al. 2025), tidybayes v3.0.7 (Kay 2024) and bayesplot v1.14.0 (Gabry et al. 2019). Posterior probability distributions for each parameter were estimated based on 8×10^4 Markov chain Monte Carlo samples (10^4 iterations $\times$ 8 chains). Model performance was assessed using $\hat{R}$ scores and trace rank and pair plots (Gabry et al. 2019). For data wrangling, summary and visualisation I used tidyverse v2.0.0 (Wickham et al. 2019), here v1.0.2 (Müller 2025), ggriidges v0.5.7 (Wilke 2021), ggh4x v0.3.1 (van den Brand 2025), geomtextpath v0.2.0 (Cameron & van den Brand 2022), patchwork v1.3.2 (Pedersen 2025) and officer v0.7.0 (Gohel et al. 2025). For details, see data and code at github.com/lukaseamus/plant-limbo. Vector graphics were polished in Affinity Designer v1.10.8 (Serif Ltd., Nottingham, UK).

To contextualise μ (days), I formally compared it with the half-life $t_{1/2}$ (days), its equivalent in the exponential model of decomposition. To facilitate the comparison, I hereinafter refer to μ as the photosynthetic half-life and $t_{1/2}$ as the detrital half-life. $t_{1/2}$ is related to the exponential decay constant k by $t_{1/2} = \ln 2 / k$ (Olson 1963). k is usually estimated using the exponential decay function but can also be approximated by the detrital turnover rate (P/B), the ratio of detrital production to the detrital pool (Cebrián & Lartigue 2004). Using published datasets (Cebrián & Lartigue 2004, Anderson-Teixeira et al. 2018, Filbee-Dexter et al. 2024, Kennedy & Blain 2025, White & Norkko 2025, Wright 2026a), I thus extracted or calculated k for terrestrial plants, freshwater plants, seagrasses and seaweeds, and converted it to $t_{1/2}$. I then modelled $t_{1/2}$ with a log-normal multilevel intercept model (see github.com/lukaseamus/plant-limbo for details) and compared its predictions to those for μ from Equation 2. I formalised this comparison as the ratio

$\mu/t_{1/2}$, a measure of physiological influence on decomposition. $\mu/t_{1/2} = 1$ implies that detrital photosynthesis reaches half its initial value at the same time as detrital mass. If $\mu/t_{1/2} < 1$, photosynthesis decays faster than mass and is unlikely to influence decomposition. $\mu/t_{1/2} > 1$ is theoretically impossible because there can be no photosynthesis without mass, but in practice it will occur due to empirical uncertainty. $\mu/t_{1/2} > 1$ may be interpreted as strong indication that photosynthesis determines decomposition. $\mu/t_{1/2}$ was calculated for each plant group from the independently estimated posterior probability distributions for μ and $t_{1/2}$. This approach propagates all uncertainty and allows calculation of the probability that the ratio falls above or below 1.

Results

Seagrasses

Halophila ovalis photosynthesised for longer post-excision than *Amphibolis antarctica*. After 34 days, the duration of the experiment, all excised *H. ovalis* leaves were still obviously net autotrophic while all excised *A. antarctica* leaves were apparently dead. On a mass basis, *H. ovalis* displayed similar baseline light-saturated net photosynthesis ($P_{\max}$) to *A. antarctica*, but its photosynthetic half-life (μ in Equation 1) was 3.9 ± 1.6 (mean $\pm$ standard deviation) times as long (Figure 1, Table 1). On a leaf basis, baseline $P_{\max}$ of *A. antarctica* was naturally much greater but the photosynthetic half-life of *H. ovalis* was 6.8 ± 2.5 times as long (Table 1).

Mean leaf blotted mass of *H. ovalis* (initially 0.052 ± 0.005 g leaf⁻¹) declined at linear rates of 0.58 ± 0.26 mg leaf⁻¹ day⁻¹ or 1.1 ± 0.43 % day⁻¹ while that of *A. antarctica* (initially 0.64 ± 0.053 g leaf⁻¹) only declined at 2 ± 2.7 mg leaf⁻¹ day⁻¹ or 0.3 ± 0.41 % day⁻¹. Hence it is surprising that excised *H. ovalis* leaves photosynthesised for so much longer. I observed periphyton colonising *H. ovalis* but not *A. antarctica*. If periphyton contributed to photosynthesis, it may have exacerbated the observed difference in photosynthetic persistence between the two seagrasses. Accounting for the uncertainty introduced by these limitations, I expect the average seagrass to have a photosynthetic half-life of more than a week (*A. antarctica*) but less than a month (*H. ovalis*).

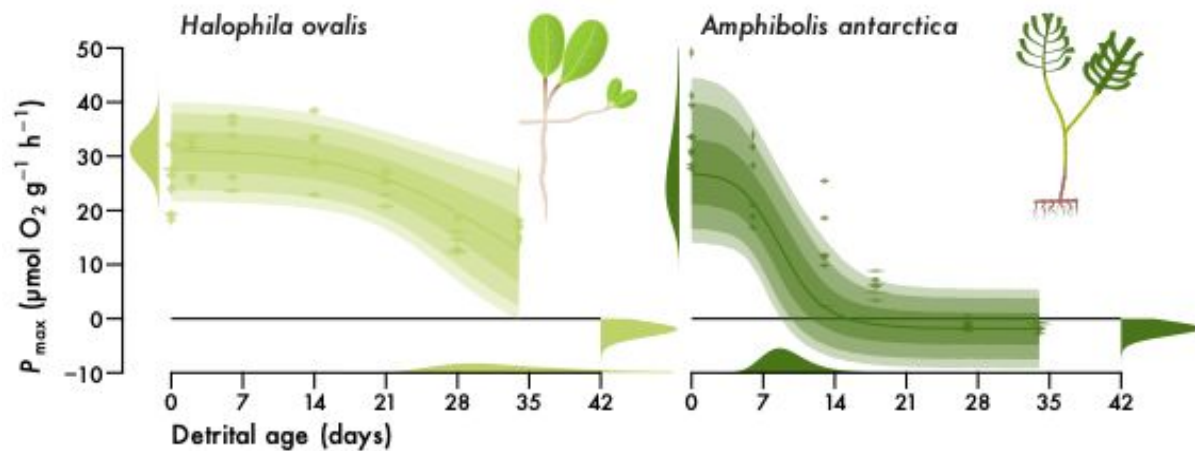


Figure 1. Seagrass detrital photosynthesis. Light-saturated net photosynthesis ($P_{\max}$) of excised *Halophila ovalis* and *Amphibolis antarctica* leaves over 34 days post-excision. $P_{\max}$ is standardised by leaf fresh mass. Violins are kernel density estimates of observations with measurement error ($n = 42$ for *H. ovalis*, $n = 39$ for *A. antarctica*). Lines and intervals are medians and the central 50, 80 and 90% of posterior probability for predicted observations. Distributions are posteriors for initial $P_{\max}$ (α), photosynthetic half-life (μ) and final respiration (τ) correctly mapped onto the axes (cf. Equation 1).

Table 1. Seagrass parameter estimates (mean $\pm$ standard deviation) for initial light-saturated net photosynthesis (α), photosynthetic half-life (μ) and final respiration (τ). Δ is the absolute difference between estimates for *H. ovalis* ($n = 42$) and *A. antarctica* ($n = 39$). $P(\Delta > 0)$ is the probability that estimates for *H. ovalis* and *A. antarctica* are different. $P = 0.5$ is equivalent to a coin toss.

Parameter	<i>Halophila ovalis</i>	<i>Amphibolis antarctica</i>	Δ	$P(\Delta > 0)$
Mass-based				
α ($\mu\text{mol g}^{-1} \text{h}^{-1}$)	31 ± 3.4	28 ± 8.4	3.3 ± 9.1	0.69
μ (days)	35 ± 11	9.5 ± 2.6	26 ± 11	1
τ ($\mu\text{mol g}^{-1} \text{h}^{-1}$)	2.1 ± 0.85	1.9 ± 0.76	0.17 ± 1.1	0.56
Leaf-based				
α ($\mu\text{mol leaf}^{-1} \text{h}^{-1}$)	1.6 ± 0.17	33 ± 12	31 ± 12	1
μ (days)	46 ± 15	7 ± 1.3	39 ± 15	1
τ ($\mu\text{mol leaf}^{-1} \text{h}^{-1}$)	0.92 ± 0.38	0.93 ± 0.37	0.0049 ± 0.53	0.51

Meta-analysis

Comparison of detrital photosynthesis across various plants revealed that light ameliorates photosynthetic decay. Detritus of terrestrial plants, seagrasses and seaweeds with access to light is predicted to have a photosynthetic half-life that is 1.8, 1.6 and 2 times as long as that in darkness (Figure 2, Table 2). Freshwater plants form an exception with a mere 68% chance (only somewhat above 50:50) of extended photosynthesis under light (Figure 2, Table 2). The predicted median light–dark ratio for unobserved plant groups is therefore 1.5 ($\ln \text{ratio} = 0.4 \pm 0.52$). In absolute terms, however, the light effect is only substantial for seaweeds (Figure 2). While median terrestrial photosynthetic half-life is only expected to increase from 7.2 to 13 days with access to light, that of seaweeds increases from 24 to 51 days (Table S2). In darkness, seaweed detritus thus only remains viable for comparable durations to other plants (Figure 2, Table S2).

The choice of response variable also matters when estimating detrital photosynthesis. Chlorophyll generally decays slower than photosynthesis. This pattern is consistent across all plants with photosynthetic half-life predicted to be 1.4 times ($\ln \text{ratio} = 0.33 \pm 0.36$) as long for chlorophyll as for photosynthesis (Figure 2, Table 2). In absolute terms, the effect again naturally dominates for seaweeds (Figure 2), with an increase from 51 days for photosynthesis to 81 days for chlorophyll under light (Table S2). Measurements based on chlorophyll therefore tend to overestimate the longevity of detrital photosynthesis.

Across light conditions and response variables, the median half-life of detrital photosynthesis ranges from 11 days for terrestrial plants, over 14 and 17 days for freshwater plants and seagrasses to 44 days for seaweeds (Table 2). The prediction for unobserved plant groups is 17 days ($\ln = 2.8 \pm 1.1$). In relative terms, seaweed detrital photosynthesis persists 3.9, 3.1, and 2.5 times as long as that of terrestrial plants, freshwater plants and seagrasses (Table 3, Figure S2). All vascular plant detritus remains viable for similar durations but there is a tendency for that of freshwater plants to have a somewhat longer photosynthetic half-life compared to terrestrial plants, with that of seagrasses being longer still (Table 3, Figure S2).

Table 2. Meta-analysis parameter estimates (Equation 2) for plant groups. Photosynthetic half-life (μ) is predicted for new experiments on unobserved species across the light (β) and

chlorophyll (γ) effects. Exponentiating β and γ yields relative effect sizes (cf. Figure 2). μ is also expressed relative to detrital half-life ($t_{1/2}$) to quantify the physiological influence on decomposition (cf. Figure 3). All parameter estimates are given as median (ln mean $\pm$ standard deviation) to accurately represent right-skewed probability distributions. n and P denote sample size and probability.

Parameter	Terrestrial plants	Freshwater plants	Seagrasses	Seaweeds
n	7240	673	160	2573
μ (days)	11 (2.4 ± 0.92)	14 (2.7 ± 0.95)	17 (2.9 ± 0.97)	44 (3.8 ± 0.96)
e^β	1.8 (0.61 ± 0.18)	1.2 (0.16 ± 0.37)	1.6 (0.47 ± 0.4)	2 (0.77 ± 0.47)
$P(\beta > 0)$	1	0.68	0.89	0.98
e^γ	1.3 (0.29 ± 0.15)	1.5 (0.41 ± 0.26)	1.4 (0.32 ± 0.3)	1.5 (0.47 ± 0.29)
$P(\gamma > 0)$	0.97	0.96	0.89	0.98
$t_{1/2}$ (days)	409 (6 ± 1.5)	57 (4 ± 1.5)	68 (4.2 ± 1.5)	14 (2.7 ± 1.5)
$\mu/t_{1/2}$	0.028 (-3.6 ± 1.8)	0.25 (-1.4 ± 1.8)	0.25 (-1.4 ± 1.8)	3 (1.1 ± 1.8)
$P(\mu/t_{1/2} > 1)$	0.02	0.22	0.23	0.73

Notwithstanding these general trends, there is substantial interspecific and experimental variation in photosynthetic half-life (Figure 3). Median estimates for species range from 3–4 days for the wheat *Triticum aestivum* to 154 days for the Antarctic brown seaweed *Desmarestia anceps* (Table S3). Median estimates for experiments range from 1.2 to 438 days. The variance in photosynthetic half-life between plant groups, species and experiments is directly quantified by logarithmic partial pooling standard deviations (σ_α in Equation 2). These standard deviations respectively are 0.51 ± 0.16 , 0.53 ± 0.12 and 0.72 ± 0.057 across plant groups, species and experiments, so experimental variation is the greatest source of uncertainty.

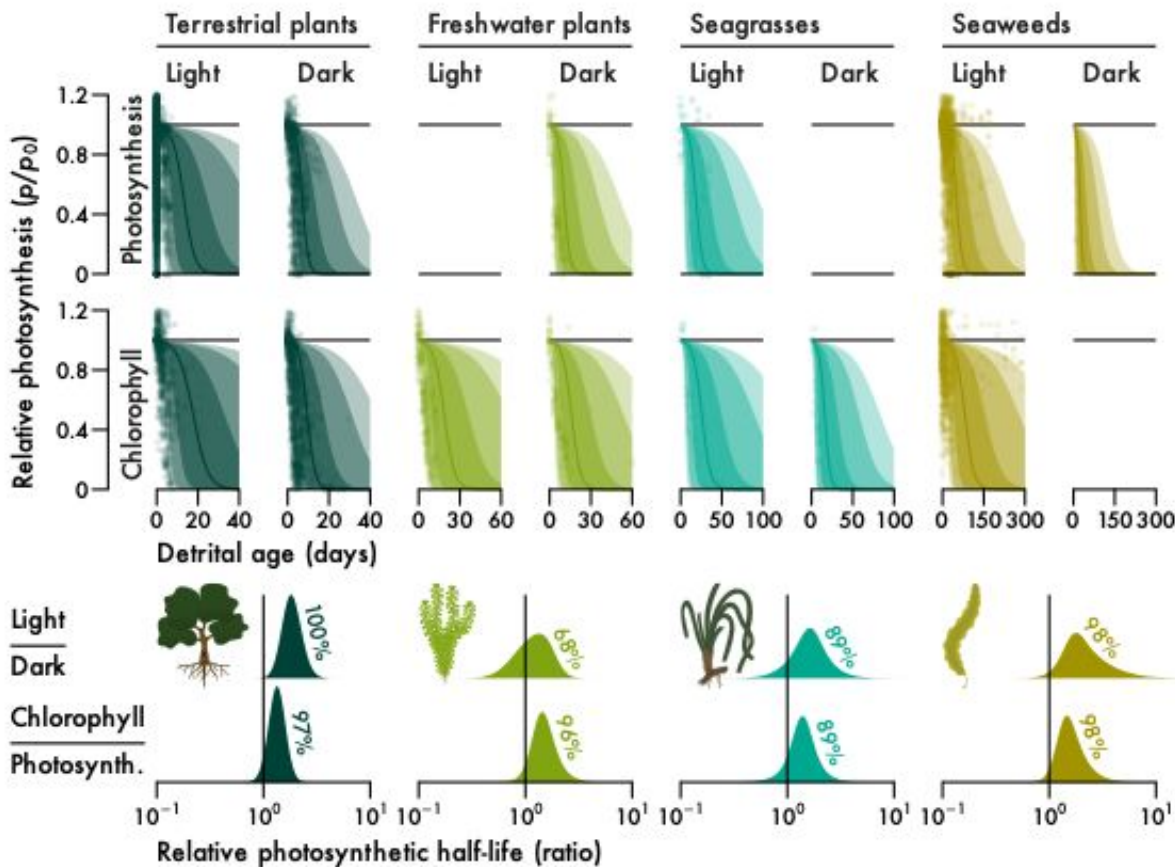


Figure 2. The effect of light and response variable type on detrital photosynthesis for major plant groups: terrestrial plants (Streptophyta excluding aquatic plants), freshwater plants (Alismatales excluding seagrasses), seagrasses, and seaweeds (Chlorophyta and Heterokontophyta excluding freshwater macroalgae). Points are observations ($n = 7240, 673, 160$ and 2573 for terrestrial plants, freshwater plants, seagrasses and seaweeds). Note that there are several observations outside the axes ranges. Lines and intervals are medians and the central 50, 80 and 90% of posterior probability for the median prediction ($e^{\bar{p}}$ in Equation 2). Note that data are missing where panels are empty and predictions are therefore not shown. Distributions are kernel density estimates of posterior probabilities for the light and chlorophyll relative effect sizes (e^{β} and e^{γ} in Equation 2). Percentages are probabilities for $\beta > 0$ and $\gamma > 0$.

Decomposition timescales also vary substantially across plant groups. Terrestrial plant detritus decomposes very slowly, only reaching half its initial mass after over a year on average (Figure 3, Table 2). Aquatic angiosperms decompose much faster, with a median detrital half-life

ranging from 57 days for freshwater plants to 68 days for seagrasses (Figure 3, Table 2), only 14% to 17% of terrestrial half-life (Table 3). Seaweeds decompose fastest, with a median detrital half-life of only two weeks (Figure 3, Table 2), 21–26% of aquatic angiosperm half-life and only 3.5% of terrestrial half-life (Table 3). Photosynthetic half-life must therefore be viewed in relation to detrital half-life to assess how affected decomposition is by detrital photosynthesis.

The ratio of photosynthetic to detrital half-life ($\mu/t_{1/2}$) quantifies the relative influence of detrital photosynthesis on decomposition, where ratios below 1 indicate little influence and ratios of 1 and above indicate substantial influence. The median of $\mu/t_{1/2}$ ranges from 0.028 for terrestrial plants over 0.25 for freshwater plants and seagrasses to 3 for seaweeds (Figure 3, Table 2). Seaweeds are therefore the only plants with a median ratio above 1 and there is a 73% chance that the ratio for any seaweed is above 1, while the probability for aquatic angiosperms is 22–23% and only 2% for terrestrial plants (Figure 3, Table 2).

Table 3. Photosynthetic (μ) and detrital ($t_{1/2}$) half-life contrasts between plant groups (cf. Figure S2). $P(\text{ratio} > 1)$ is the probability that estimates are different. $P = 0.5$ is equivalent to a coin toss. All estimates except P are given as median (ln mean $\pm$ standard deviation). See Table 2 for sample sizes.

Ratio	μ/μ	$P(\mu/\mu > 1)$	$t_{1/2}/t_{1/2}$	$P(t_{1/2}/t_{1/2} > 1)$
Seaweeds / Terrestrial	3.9 (1.4 $\pm$ 1.3)	0.85	0.035 (−3.3 $\pm$ 2.1)	0.057
Seaweeds / Freshwater	3.1 (1.1 $\pm$ 1.4)	0.80	0.26 (−1.4 $\pm$ 2.1)	0.26
Seaweeds / Seagrasses	2.5 (0.93 $\pm$ 1.4)	0.75	0.21 (−1.6 $\pm$ 2.1)	0.23
Seagrasses / Terrestrial	1.5 (0.43 $\pm$ 1.3)	0.63	0.17 (−1.8 $\pm$ 2.1)	0.2
Freshwater / Terrestrial	1.3 (0.23 $\pm$ 1.3)	0.57	0.14 (−2 $\pm$ 2.1)	0.18
Seagrasses / Freshwater	1.2 (0.2 $\pm$ 1.3)	0.56	1.2 (0.18 $\pm$ 2.2)	0.53

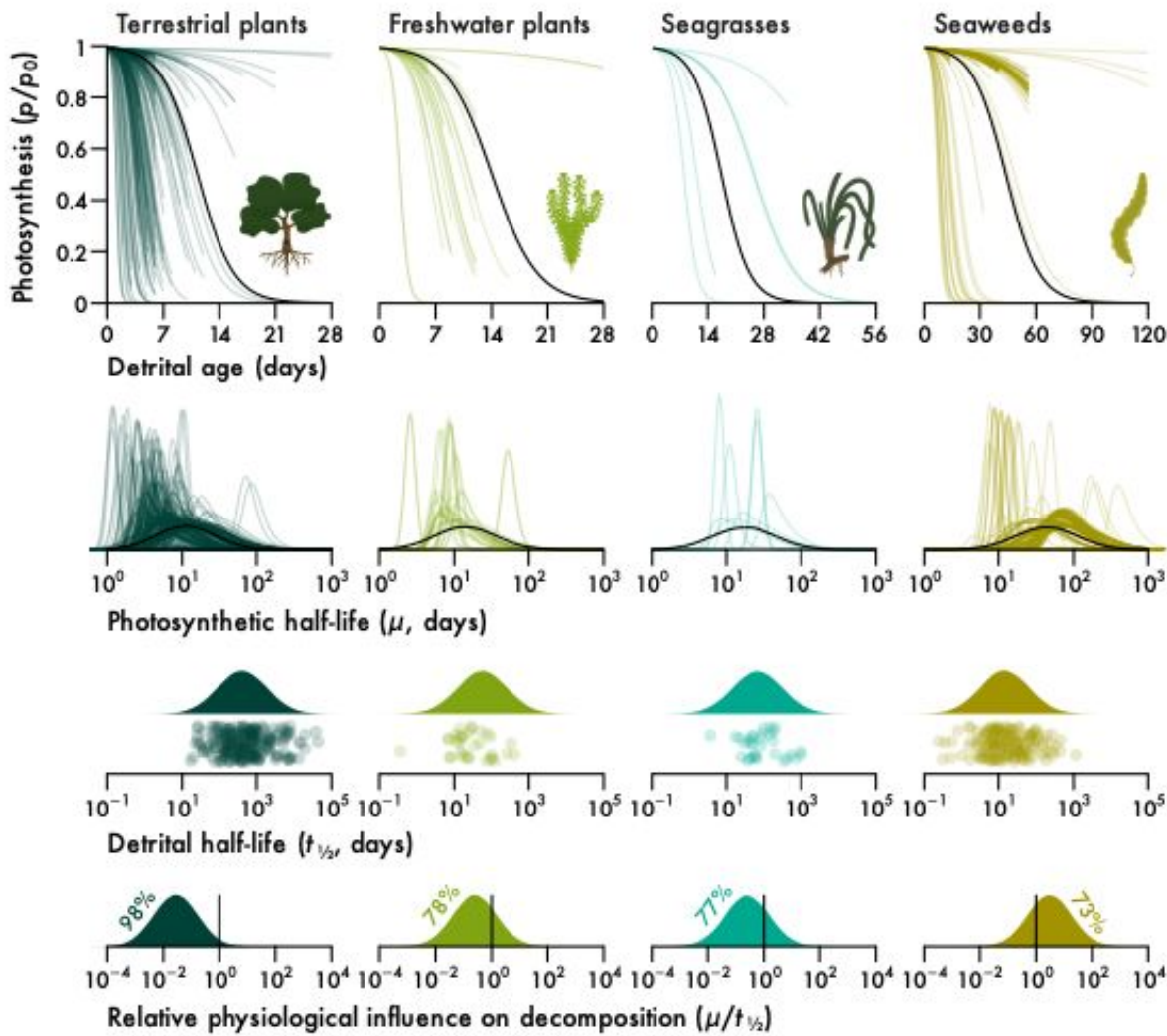


Figure 3. Longevity of detrital photosynthesis relative to detrital mass for major plant groups (cf. Figure 2). Lines are median predictions of relative photosynthesis ($e^{\bar{p}}$ in Equation 2) or posterior probability distributions of photosynthetic half-life (μ in Equation 2). Existing experiments ($n = 314, 32, 8$ and 197 for terrestrial plants, freshwater plants, seagrasses and seaweeds) are given in colour and predictions for new experiments on unobserved species across the light and chlorophyll effects are given in black. Raincloud plots of detrital half-life ($t_{1/2}$ sensu Olson 1963) show observations ($n = 239, 33, 32$ and 171 for terrestrial plants, freshwater plants, seagrasses and seaweeds) below posterior probability distributions for new observations. Distributions in the bottom panel are ratios of the black distributions of μ to the distributions of $t_{1/2}$. Percentages are probabilities for $\mu/t_{1/2} < 1$ or $\mu/t_{1/2} > 1$, depending on what side of the identity line most of the probability lies.

Discussion

Some plant detritus can maintain photosynthesis for long periods following detachment. This is at odds with conventional understanding of detritus as dead organic matter. Despite the existence of various relevant literature published on this topic since the 1950s and recognition of the phenomenon in seaweeds since the 1980s, detrital photosynthesis had not been explored beyond kelps or formally conceptualised. In part, this is probably due to diverse research objectives and terminology which obscure the connection between quantitatively comparable data. There was also data scarcity for seagrasses which, as the only vascular counterpart to seaweeds, form an important functional link to freshwater and terrestrial plants. Here, I collected the first data on photosynthesis in seagrass detritus, developed a mathematical model of detrital photosynthesis and used it to make the first quantitative comparison of photosynthetic persistence in the detrital phase of various plants. Results suggest that the non-vascular macroalgae have the greatest potential for detrital photosynthesis while among the vascular plants there is slightly increased detrital viability with a shift from a terrestrial to an aquatic and yet again to a marine lifestyle. Although all plants display detrital photosynthesis to some extent, it does not necessarily influence their decomposition. Terrestrial plant decomposition is slower than that of aquatic vascular plants which in turn is much slower than that of macroalgae. I predict that detrital photosynthesis is a general phenomenon but its effect on decomposition ranges from negligible for terrestrial plants, over measurable for aquatic vascular plants, to substantial for macroalgae. It is conceivable that detrital photosynthesis determines macroalgal decomposition. I therefore propose a new causal model of macroalgal decomposition that emerges from my formalisation of the theory of detrital photosynthesis (Figure 4). While conventional decomposition theory may continue to be applied to vascular plants and detritus *sensu stricto*, its assumptions will not always hold for aquatic systems and certainly do not apply to macroalgae (Figure 4).

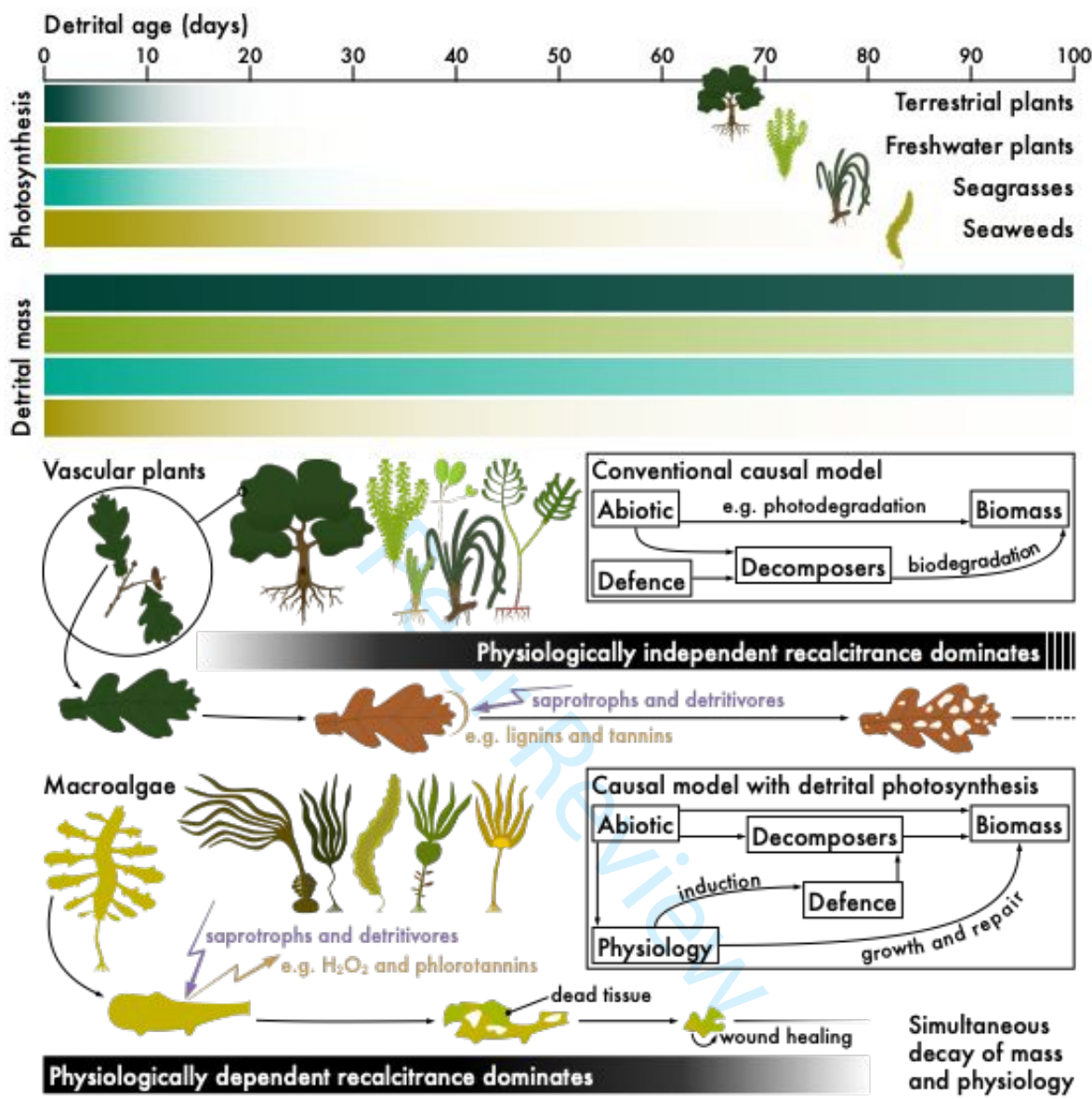


Figure 4. Conceptual model of decomposition in vascular plants and macroalgae. Coloured bars show the decay of detrital photosynthesis and biomass for major plant groups, based respectively on the median for the posterior prediction of relative photosynthesis p (Equation 2) and e^{-kt} where the decay rate k is derived from detrital half-life ($t_{1/2}$) as the median of $\ln 2 / t_{1/2}$ (cf. Figure 3). Black bars show approximate timelines for the dominance of physiologically independent and dependent resistance to decomposition (i.e. recalcitrance) in vascular plants and macroalgae respectively. Insets show the conventional and new causal models as directed acyclic graphs (McElreath 2020) where nodes are variables and arrows show their causal relationships. Some arrows are labelled with mechanisms for clarity. Illustrations connected by arrows show the

schematic decomposition process over time. Note that some illustrated concepts are speculative and further empirical work is needed to validate them (see conclusion).

Detrital photosynthesis per se is not restricted to non-vascular plants. Seagrass detritus can also photosynthesise for at least a week. The oxygen production of the diverse seagrasses *Halophila ovalis* and *Amphibolis antarctica* responds similarly to detrital age and the delayed response in *H. ovalis* may simply be attributable to greater colonisation by periphyton. Given a leaf mass per area of 2.7 and 6.8 mg cm⁻² for *H. ovalis* and *A. antarctica* respectively (de los Santos et al. 2012, 2016), periphyton could contribute 12–172% of initial seagrass photosynthesis (Neely & Wetzel 1997). Alternatively, *A. antarctica* leaves may be more reliant on their roots for nutrient provisioning, although there is currently no evidence to suggest this (Pedersen et al. 1997). The 26–39-day shorter photosynthetic half-life of isolated *A. antarctica* leaves compared to *H. ovalis* is surprising given their 1.7 times or 47-day longer attached lifespan (Hemminga et al. 1999). Nonetheless, *Halophila* detritus is often viable (Hall et al. 2006, Weatherall et al. 2025) and can continue growing to a similar or greater extent relative to other seagrasses (Hanuise et al. 2026). Despite observations of seagrass detrital viability (Hall et al. 2006, Stafford-Bell et al. 2015, Weatherall et al. 2025) and growth (Hanuise et al. 2026) for up to a month and my evidence of photosynthetic persistence for a similar duration, seagrass detrital photosynthesis may not necessarily counteract decomposition. I found that seagrass detritus has a detrital half-life of 68 days on average and it may take a year for the majority to decompose (Harrison 1989, Cebrián et al. 1997, Cebrián & Lartigue 2004, Trevathan-Tackett et al. 2020). This is much longer than detached seagrass is expected to photosynthesise and accordingly there is little evidence of physiological influence on its decomposition. While live *Halophila decipiens* ramets do indeed decompose slower than buried ones (Kenworthy et al. 1989), decomposition rates are similar between live and senescent or buried *Thalassia testudinum* leaves (Ruble & Roman 1982, Fourqurean & Schrlau 2003, Rosch & Koch 2009) and reviews suggest that live seagrass leaves actually tend to decompose faster than senescent or dead ones (Harrison 1989, Trevathan-Tackett et al. 2020).

Detrital photosynthesis is favoured primarily by non-vascularity and secondarily by an aquatic lifestyle. Macroalgae uniquely exhibit long-term detrital photosynthesis which clearly sets them

apart from vascular plants (Figure 4). Nevertheless, seagrasses and freshwater plants on average display detrital photosynthesis several days longer than terrestrial plants. This is almost certainly due to the greater photosynthetic independence of aquatic plant leaves. Because most experiments are carried out in water, they probably overestimate the persistence of detrital photosynthesis in terrestrial plants. With access to light but not water, which is the usual fate of terrestrial plant detritus, detached leaves of various trees tend to lose the ability to photosynthesise in just a few minutes (e.g. Gauthier & Jacobs 2018, Kar et al. 2021) due to breakdown of stomatal conductance following loss of water pressure (Powles et al. 2006, Gauthier & Jacobs 2018). The true difference between aquatic and terrestrial vascular plants may therefore be greater than predicted by my meta-analysis. Surprisingly, seagrasses detritus tends to photosynthesise for longer than that of freshwater plants from the same order. The adoption of a marine lifestyle and convergent evolution with seaweeds (Olsen et al. 2016) may be the ultimate reason. But a more proximate explanation is excess production of hydrogen peroxide in detached freshwater plant leaves (Jana & Choudhuri 1982, Kar & Choudhuri 1986, 1987) which may be absent in seagrass leaves. As demonstrated for the water thyme *Hydrilla verticillata* (Kar & Choudhuri 1986, 1987), hydrogen peroxide production is exacerbated by photosynthesis, leading to light-induced chlorophyll degradation that is independent of photodegradation. Hence freshwater plants were the only group for which I did not find a positive light effect on photosynthetic half-life. In contrast, there is no evidence for increased chlorophyll breakdown in isolated seagrass leaves exposed to light (Strother & Vatta 1986) and seagrass photodegradation has only been reported for dead leaves (Vähätalo et al. 1998). Perhaps the answer lies in alterations to the light harvesting complexes and cell wall features of seagrass leaves that enhance gas exchange (Olsen et al. 2016). Regardless, emancipation of leaf photosynthesis from the rest of the plant is apparently even more favoured by a marine than an aquatic lifestyle.

Macroalgal decomposition is uniquely determined by detrital photosynthesis. Of all considered plants, terrestrial ones decompose slowest and macroalgae decompose fastest, so the observed divergence in photosynthetic decay is increased when viewing it relative to decomposition. Macroalgae are the only plants whose detritus may photosynthesise for as long as it remains intact, aquatic vascular plants maintain photosynthesis for a quarter of their detrital phase and terrestrial decomposition is probably unaffected by physiology due to a mismatch between rapid

photosynthetic decay and slow decomposition (Figure 4). A clear distinction must here be made between vascular plants and macroalgae. Since the majority of vascular plant decomposition occurs in the absence of photosynthesis, its main intrinsic predictor must be physiologically independent constitutive defences such as lignins and tannins, while that of macroalgal decomposition is likely physiologically induced resistance to decay (Figure 4). Powered by photosynthesis, detrital physiology may be able to respond to decomposition with growth, repair and chemical defence (Figure 4). These mechanisms can go hand in hand as in the case of phlorotannins acting both as repair and defence in excised brown seaweeds (Lüder & Clayton 2004, Halm et al. 2011). While there is evidence for the persistence and increase of detrital phlorotannins (Tala et al. 2013, 2016, 2019, de Bettignies et al. 2020, Wright et al. 2022, but see Pedersen et al. 2021), other common chemical defences such as induced (Weinberger et al. 1999, Weinberger & Friedlander 2000, Küpper et al. 2001, 2002, Thomas et al. 2014) and constitutive (Küpper et al. 2002, Twigg et al. 2020) hydrogen peroxide release have yet to be measured in detritus. Nonetheless, the presented evidence for detrital photosynthesis clearly introduces novel causal paths (Figure 4). For example, light prolongs detrital photosynthesis which may slow macroalgal decomposition through growth and provide a mechanism contrary to photodegradation. In this case the mechanism is clear, but the same outcome can also be produced by different mechanisms (Figure 4). For instance, elevated temperature may accelerate decomposition in three ways: by reducing the structural integrity of detritus, by increasing decomposer metabolism, and also by diminishing detrital photosynthesis as was recently demonstrated (Wright et al. 2024). Further research is required to tease apart these causal relationships.

Detrital photosynthesis should be considered when making inferences about macroalgal decomposition. Firstly, the initial physiological state of detritus in a decomposition experiment may not reflect that in nature. In contrast to seagrass litter bag experiments (Harrison 1989, Trevathan-Tackett et al. 2020), most seaweed decomposition experiments have only used freshly excised blades as a detritus proxy (Frontier et al. 2021, Filbee-Dexter et al. 2022, Frontier et al. 2022, Wright et al. 2022, Kennedy & Blain 2025, White & Norkko 2025). Senescent or naturally detached fronds are rarely included (Hamersley et al. 2015, de Bettignies et al. 2020) and the practice of pre-killing tissue to obtain detritus *sensu stricto* is restricted to few, mostly early

studies (Josselyn & Mathieson 1980, Rice & Tenore 1981, Birch et al. 1983, Smith & Foreman 1984, Brouwer 1996). While the fraction of seaweed detritus that consists of whole dislodged plants or fronds is certainly initially viable, the majority of tissue that is abscised or eroded likely exhibits impaired photosynthesis from the start (Küppers & Kremer 1978, Arnold & Manley 1985, Gómez et al. 1996, Rodriguez et al. 2016). Ignorance of the physiology of experimental tissues on the one hand leads to the erroneous conclusion that decomposition rates derived from freshly excised seaweeds are representative of global carbon cycling (e.g. Filbee-Dexter et al. 2024), and on the other hand results in practices that inadvertently impair photosynthesis to an unknown degree, such as long sample transport (e.g. Pedersen et al. 2021). Secondly, conditions during a decomposition experiment may have unintended effects on detrital photosynthesis which can then be misattributed. As I have shown, detrital photosynthesis is highly sensitive to light in most plants (Figure 2) and temperature in kelps (Wright et al. 2024). Oxygen is undoubtedly also important and there are probably many more nuanced physiological stressors that remain overlooked. For example, decomposition rates measured under any influence of even weak light will be highly misleading when the goal is to predict carbon cycling in the deep ocean (Filbee-Dexter et al. 2024, White & Norkko 2025). Or the accelerating effect of temperature on decomposition may be attributed to structural weakening of detritus and elevated microbial metabolism rather than impaired detrital photosynthesis (Filbee-Dexter et al. 2022, Frontier et al. 2022). These confounders are not immediately obvious without the causal link provided by detrital physiology (Figure 4). Detrital photosynthesis likely accounts for the large unexplained variation in macroalgal decomposition rates observed across experiments (Kennedy & Blain 2025, White & Norkko 2025). Awareness of the causation revealed by the presented theory (Figure 4) is necessary to properly design and interpret decomposition experiments involving macroalgae and perhaps aquatic plants in general.

To conclude, I show that all plants may photosynthesise following detachment, but timescales vary greatly from a few minutes in tree leaves without access to water to a year for certain seaweeds. Only macroalgal detrital photosynthesis decays on similar timescales to detrital mass. Physiology is therefore expected to substantially influence the decomposition of macroalgae, a causal relationship which is unique among plants. The initial physiological state of detritus and predictors of photosynthesis, such as light and temperature, can hence aid inference of

macroalgal decomposition. With the theory of detrital photosynthesis formalised, we now need (1) a mathematical model of macroalgal decomposition that takes this theory into account — I have recently made some progress in this direction (Wright 2026b), (2) more decomposition experiments comparing live with dead detritus (cf. Birch et al. 1983, Brouwer 1996) or fresh with senescent tissues (Hamersley et al. 2015, de Bettignies et al. 2020) to corroborate the proposed effect of physiology on decomposition, (3) surveys of the initial physiological state of detritus produced by erosion, abscission and dislodgement, (4) in situ measurements of detrital photosynthesis in naturally deposited detrital accumulations, (5) further studies on post-excision photosynthesis, especially in underrepresented plants such as seagrasses and seaweeds other than kelps, and for missing treatment combinations such as freshwater plants in light and seagrasses in darkness (Figure 2), and (6) investigation into other aspects of detrital physiology such as induced chemical defence.

Acknowledgements

Thank you to Andy Foggo and Nadia Frontier for inspiring discussions over the years, Belinda Martin for introducing me to *Halophila ovalis* and joining me in the field, Yannick Mulders and Thomas Wernberg for supplying and Chunbo Liu for repairing the oxygen meter, and Nadia Frontier and Dan van Hees for sharing data.

Financial support

This work was supported by the Forrest Research Foundation Scholarship and University of Western Australia International Fee Scholarship and University Postgraduate Award.

Conflicts of interest

I declare no conflicts of interest.

Data and code availability

Data and annotated code are available at github.com/lukaseamus/plant-limbo. The dataset used for meta-analysis (Wright 2025) can also be accessed at github.com/lukaseamus/detrital-photosynthesis.

References

- Anderson-Teixeira KJ, Wang MMH, McGarvey JC, Herrmann V, Tepley AJ, Bond-Lamberty B, LeBauer DS (2018) ForC: a global database of forest carbon stocks and fluxes. *Ecology* 99, 1507. <https://doi.org/10.1002/ecy.2229>.
- Arnold KE, Manley SL (1985) Carbon allocation in *Macrocystis pyrifera* (Phaeophyta): intrinsic variability in photosynthesis and respiration. *Journal of Phycology* 21, 154–167. <https://doi.org/10.1111/j.0022-3646.1985.00154.x>.
- Birch PB, Gabrielson JO, Hamel KS (1983) Decomposition of *Cladophora*. I. Field studies in the Peel-Harvey estuarine system, Western Australia. *Botanica Marina* 26, 165–172. <https://doi.org/10.1515/botm.1983.26.4.165>.
- Bolton JJ (2016) What is aquatic botany? — and why algae are plants: the importance of non-taxonomic terms for groups of organisms. *Aquatic Botany* 132, 1–4. <https://doi.org/10.1016/j.aquabot.2016.02.006>.
- Bourguès S, Auby I, de Wit R, Labourg P-J (1996) Differential anaerobic decomposition of seagrass (*Zostera noltii*) and macroalgal (*Monostroma obscuro*) biomass from Arcachon Bay (France). *Hydrobiologia* 329, 121–131. <https://doi.org/10.1007/bf00034552>.
- Brouwer PEM (1996) Decomposition in situ of the sublittoral Antarctic macroalga *Desmarestia anceps* Montagne. *Polar Biology* 16, 129–137. <https://doi.org/10.1007/bf02390433>.
- Cameron A, van den Brand T (2022) geomtextpath: use curved text in ggplot2. <https://allanccameron.github.io/geomtextpath>.
- Carpenter B, Gelman A, Hoffman MD, Lee D, Goodrich B, Betancourt M, Brubaker M, Guo J, Li P, Riddell A (2017) Stan: a probabilistic programming language. *Journal of Statistical Software* 76, 1–32. <https://doi.org/10.18637/jss.v076.i01>.
- Cebrián J, Duarte CM, Marbà N, Enríquez S (1997) Magnitude and fate of the production of four co-occurring Western Mediterranean seagrass species. *Marine Ecology Progress Series* 155, 29–44. <https://doi.org/10.3354/meps155029>.
- Cebrián J, Lartigue J (2004) Patterns of herbivory and decomposition in aquatic and terrestrial ecosystems. *Ecological Monographs* 74, 237–259. <https://doi.org/10.1890/03-4019>.
- de Bettignies F, Dauby P, Thomas F, Gobet A, Delage L, Böhner O, Loisel S, Davoult D (2020) Degradation dynamics and processes associated with the accumulation of *Laminaria*

- 693 *hyperborea* (Phaeophyceae) kelp fragments: an in situ experimental approach. *Journal of*
 694 *Phycology* 56, 1481–1492. <https://doi.org/10.1111/jpy.13041>.
- 695 de los Santos CB, Onoda Y, Vergara JJ, Pérez-Lloréns JL, Bouma TJ, La Nafie YA, Cambridge
 696 ML, Brun FG (2016) A comprehensive analysis of mechanical and morphological traits in
 697 temperate and tropical seagrass species. *Marine Ecology Progress Series* 551, 81–94.
 698 <https://doi.org/10.3354/meps11717>.
- 699 de los Santos CB, Brun FG, Onoda Y, Cambridge ML, Bouma TJ, Vergara JJ, Pérez-Lloréns JL
 700 (2012) Leaf-fracture properties correlated with nutritional traits in nine Australian seagrass
 701 species: implications for susceptibility to herbivory. *Marine Ecology Progress Series* 458,
 702 89–102. <https://doi.org/10.3354/meps09757>.
- 703 Drew EA (1983) Physiology of *Laminaria*. I. Use of excised lamina discs in short and long term
 704 experiments. *Marine Ecology* 226, 259–270. [https://doi.org/10.1111/j.1439-](https://doi.org/10.1111/j.1439-0485.1983.tb00297.x)
 705 [0485.1983.tb00297.x](https://doi.org/10.1111/j.1439-0485.1983.tb00297.x).
- 706 Evrard V, Kiswara W, Bouma TJ, Middelburg JJ (2005) Nutrient dynamics of seagrass
 707 ecosystems: ¹⁵N evidence for the importance of particulate organic matter and root systems.
 708 *Marine Ecology Progress Series* 295, 49–55. <https://doi.org/10.3354/meps295049>.
- 709 Filbee-Dexter K, Feehan CJ, Smale DA, Krumhansl KA, Augustine S, de Bettignies F, Burrows
 710 MT, Byrnes JEK, Campbell J, Davoult D et al. (2022) Kelp carbon sink potential decreases
 711 with warming due to accelerating decomposition. *PLoS Biology* 20, e3001702.
 712 <https://doi.org/10.1371/journal.pbio.3001702>.
- 713 Filbee-Dexter K, Pessarrodona A, Pedersen MF, Wernberg T, Duarte CM, Assis J, Bekkby T,
 714 Burrows MT, Carlson DF, Gattuso J-P (2024) Carbon export from seaweed forests to deep
 715 ocean sinks. *Nature Geoscience* 17, 552–559. <https://doi.org/10.1038/s41561-024-01449-7>.
- 716 Fourqurean JW, Schrlau JE (2003) Changes in nutrient content and stable isotope ratios of C and
 717 N during decomposition of seagrasses and mangrove leaves along a nutrient availability
 718 gradient in Florida Bay, USA. *Chemistry and Ecology* 19, 373–390.
 719 <https://doi.org/10.1080/02757540310001609370>.
- 720 Fraser CI, Morrison AK, Hogg AM, Macaya EC, van Sebille E, Ryan PG, Padovan A, Jack C,
 721 Valdivia N, Waters JM (2018) Antarctica’s ecological isolation will be broken by storm-
 722 driven dispersal and warming. *Nature Climate Change* 8, 704–708.
 723 <https://doi.org/10.1038/s41558-018-0209-7>.

- 724 Frontier N, de Bettignies F, Foggo A, Davoult D (2021) Sustained productivity and respiration of
725 degrading kelp detritus in the shallow benthos: detached or broken, but not dead. *Marine*
726 *Environmental Research* 166, 105277. <https://doi.org/10.1016/j.marenvres.2021.105277>.
- 727 Frontier N, Mulas M, Foggo A, Smale DA (2022) The influence of light and temperature on
728 detritus degradation rates for kelp species with contrasting thermal affinities. *Marine*
729 *Environmental Research* 173, 105529. <https://doi.org/10.1016/j.marenvres.2021.105529>.
- 730 Gabry J, Češnovar R, Johnson A, Bronder S (2025) cmdstanr: R Interface to ‘CmdStan’.
731 <https://mc-stan.org/cmdstanr>.
- 732 Gabry J, Simpson D, Vehtari A, Betancourt M, Gelman A (2019) Visualization in Bayesian
733 workflow. *Journal of the Royal Statistical Society Series A: Statistics in Society* 182, 389–
734 402. <https://doi.org/10.1111/rssa.12378>.
- 735 Gauthier M-M, Jacobs DF (2018) Reductions in net photosynthesis and stomatal conductance
736 vary with time since leaf detachment in three deciduous angiosperms. *Trees* 32, 1247–
737 1252. <https://doi.org/10.1007/s00468-018-1706-z>.
- 738 Gohel D, Moog S, Heckmann M (2025) officer: manipulation of Microsoft Word and
739 PowerPoint documents. <https://ardata-fr.github.io/officeverse>.
- 740 Gómez I, Wiencke C, Thomas DN (1996) Variations in photosynthetic characteristics of the
741 Antarctic marine brown alga *Ascoseira mirabilis* in relation to thallus age and size.
742 *European Journal of Phycology* 31, 167–172. <https://doi.org/10.1080/09670269600651341>.
- 743 Graiff A, Karsten U, Meyer S, Pfender D, Tala F, Thiel M (2013) Seasonal variation in floating
744 persistence of detached *Durvillaea antarctica* (Chamisso) Hariot thalli. *Botanica Marina*
745 56, 3–14. <https://doi.org/10.1515/bot-2012-0193>.
- 746 Graiff A, Pantoja JF, Tala F, Thiel M (2016) Epibiont load causes sinking of viable kelp rafts:
747 seasonal variation in floating persistence of giant kelp *Macrocystis pyrifera*. *Marine*
748 *Biology* 163, 191. <https://doi.org/10.1007/s00227-016-2962-3>.
- 749 Gras AF, Koch MS, Madden CJ (2003) Phosphorus uptake kinetics of a dominant tropical
750 seagrass *Thalassia testudinum*. *Aquatic Botany* 76, 299–315. [https://doi.org/10.1016/s0304-](https://doi.org/10.1016/s0304-3770(03)00069-x)
751 [3770\(03\)00069-x](https://doi.org/10.1016/s0304-3770(03)00069-x).
- 752 Guiry, M.D., Guiry GM (2025) AlgaeBase. <https://www.algaebase.org>.

- 753 Hall LM, Hanisak MD, Virnstein RW (2006) Fragments of the seagrasses *Halodule wrightii* and
 754 *Halophila johnsonii* as potential recruits in Indian River Lagoon, Florida. *Marine Ecology*
 755 *Progress Series* 310, 109–117. <https://doi.org/10.3354/meps310109>.
- 756 Halm H, Lüder UH, Wiencke C (2011) Induction of phlorotannins through mechanical wounding
 757 and radiation conditions in the brown macroalga *Laminaria hyperborea*. *European Journal*
 758 *of Phycology* 46, 16–26. <https://doi.org/10.1080/09670262.2010.511277>.
- 759 Hamersley MR, Sohm JA, Burns JA, Capone DG (2015) Nitrogen fixation associated with the
 760 decomposition of the giant kelp *Macrocystis pyrifera*. *Aquatic Botany* 125, 57–63.
 761 <https://doi.org/10.1016/j.aquabot.2015.05.003>.
- 762 Hanuise D, Lin C, Tol S, Choukroun S, Rasheed MA, Dobbelaere T, York PH, Smith TM, Coles
 763 RG, Hanert E et al. (2026) Experimentally derived buoyancy duration of seagrass
 764 fragments for biophysical dispersal modelling in the Great Barrier Reef. *Marine*
 765 *Environmental Research* 217, 107937. <https://doi.org/10.1016/j.marenvres.2026.107937>.
- 766 Harrison PG (1989) Detrital processing in seagrass systems: a review of factors affecting decay
 767 rates, remineralization and detritivory. *Aquatic Botany* 35, 263–288.
 768 [https://doi.org/10.1016/0304-3770\(89\)90002-8](https://doi.org/10.1016/0304-3770(89)90002-8).
- 769 Hemminga MA, Marba N, Stapel J (1999) Leaf nutrient resorption, leaf lifespan and the
 770 retention of nutrients in seagrass systems. *Aquatic Botany* 65, 141–158.
 771 [https://doi.org/10.1016/s0304-3770\(99\)00037-6](https://doi.org/10.1016/s0304-3770(99)00037-6).
- 772 Hernández-Carmona G, Hughes B, Graham MH (2006) Reproductive longevity of drifting kelp
 773 *Macrocystis pyrifera* (Phaeophyceae) in Monterey Bay, USA. *Journal of Phycology* 42,
 774 1199–1207. <https://doi.org/10.1111/j.1529-8817.2006.00290.x>.
- 775 Jana S, Choudhuri MA (1980) Senescence in submerged aquatic angiosperms: changes in intact
 776 and isolated leaves during aging. *New Phytologist* 86, 191–198.
 777 <https://doi.org/10.1111/j.1469-8137.1980.tb03188.x>.
- 778 Jana S, Choudhuri MA (1982) Glycolate metabolism of three submersed aquatic angiosperms
 779 during ageing. *Aquatic Botany* 12, 345–354. [https://doi.org/10.1016/0304-3770\(82\)90026-](https://doi.org/10.1016/0304-3770(82)90026-2)
 780 2.
- 781 Jiménez-Ramos R, Henares C, Egea LG, Vergara JJ, Brun FG (2023) Leaf senescence of the
 782 seagrass *Cymodocea nodosa* in Cádiz Bay, Southern Spain. *Diversity* 15, 187.
 783 <https://doi.org/10.3390/d15020187>.

- Josselyn MN, Mathieson AC (1980) Seasonal influx and decomposition of autochthonous macrophyte litter in a north temperate estuary. *Hydrobiologia* 71, 197–208. <https://doi.org/10.1007/bf03216236>.
- Kar RK, Choudhuri MA (1987) Possible mechanisms of light-induced chlorophyll degradation in senescing leaves of *Hydrilla verticillata*. *Physiologia Plantarum* 70, 729–734. <https://doi.org/10.1111/j.1399-3054.1987.tb04331.x>.
- Kar RK, Choudhuri MA (1986) Effects of light and spermine on senescence of *Hydrilla* and spinach leaves. *Plant Physiology* 80, 1030–1033. <https://doi.org/10.1104/pp.80.4.1030>.
- Kar S, Montague DT, Villanueva-Morales A (2021) Measurement of photosynthesis in excised leaves of ornamental trees: a novel method to estimate leaf level drought tolerance and increase experimental sample size. *Trees* 35, 889–905. <https://doi.org/10.1007/s00468-021-02088-w>.
- Kay M (2024) tidybayes: tidy data and geoms for Bayesian models. <https://doi.org/10.5281/zenodo.13770114>.
- Kennedy JR, Blain CO (2025) A systematic review of marine macroalgal degradation: toward a better understanding of macroalgal carbon sequestration potential. *Journal of Phycology* 61, 399–432. <https://doi.org/10.1111/jpy.70031>.
- Kenworthy WJ, Currin CA, Fonseca MS, Smith G (1989) Production, decomposition, and heterotrophic utilization of the seagrass *Halophila decipiens* in a submarine canyon. *Marine Ecology Progress Series* 51, 277–290. <https://doi.org/10.3354/meps051277>.
- Knauer GA, Ayers AV (1977) Changes in carbon, nitrogen, adenosine triphosphate, and chlorophyll *a* in decomposing *Thalassia testudinum* leaves. *Limnology and Oceanography* 22, 408–414. <https://doi.org/10.4319/lo.1977.22.3.0408>.
- Koivikko R, Lopenen J, Honkanen T, Jormalainen V (2005) Contents of soluble, cell-wall-bound and exuded phlorotannins in the brown alga *Fucus vesiculosus*, with implications on their ecological functions. *Journal of Chemical Ecology* 31, 195–212. <https://doi.org/10.1007/s10886-005-0984-2>.
- Küpper FC, Müller DG, Peters AF, Kloareg B, Potin P (2002) Oligoalginat recognition and oxidative burst play a key role in natural and induced resistance of sporophytes of Laminariales. *Journal of Chemical Ecology* 28, 2057–2081. <https://doi.org/10.1023/a:1020706129624>.

- 815 Küpper FC, Kloareg B, Guern J, Potin P (2001) Oligoguluronates elicit an oxidative burst in the
 816 brown algal kelp *Laminaria digitata*. *Plant Physiology* 125, 278–291.
 817 <https://doi.org/10.1104/pp.125.1.278>.
- 818 Küppers U, Kremer BP (1978) Longitudinal profiles of carbon dioxide fixation capacities in
 819 marine macroalgae. *Plant Physiology* 62, 49–53. <https://doi.org/10.1104/pp.62.1.49>.
- 820 Lee D, Stan buildbot, seantalts, Carpenter B, Morris M, Kucukelbir A, Betancourt M, Tran D,
 821 Brubaker M, Bales B et al. (2026) stan-dev/cmdstan. *Zenodo*.
 822 <https://doi.org/10.5281/zenodo.598253>.
- 823 Lee K-S, Dunton KH (1999) Inorganic nitrogen acquisition in the seagrass *Thalassia testudinum*:
 824 development of a whole-plant nitrogen budget. *Limnology and Oceanography* 44, 1204–
 825 1215. <https://doi.org/10.4319/lo.1999.44.5.1204>.
- 826 Lüder UH, Clayton MN (2004) Induction of phlorotannins in the brown macroalga *Ecklonia*
 827 *radiata* (Laminariales, Phaeophyta) in response to simulated herbivory—the first
 828 microscopic study. *Planta* 218, 928–937. <https://doi.org/10.1007/s00425-003-1176-3>.
- 829 Macaya EC, Boltana S, Hinojosa IA, Macchiavello JE, Valdivia NA, Vasquez NR, Buschmann
 830 AH, Vasquez JA, Alonso Vega JM, Thiel M (2005) Presence of sporophylls in floating
 831 kelp rafts of *Macrocystis* spp. (Phaeophyceae) along the Chilean Pacific coast. *Journal of*
 832 *Phycology* 41, 913–922. <https://doi.org/10.1111/j.1529-8817.2005.00118.x>.
- 833 Marbà N, Hemminga MA, Mateo MA, Duarte CM, Mass YEM, Terrados J, Gacia E (2002)
 834 Carbon and nitrogen translocation between seagrass ramets. *Marine Ecology Progress*
 835 *Series* 226, 287–300. <https://doi.org/10.3354/meps226287>.
- 836 McElreath R (2020) Statistical rethinking²: a Bayesian course with examples in R and Stan,
 837 second edition. Chapman and Hall/CRC. <https://doi.org/10.1201/9780429029608>.
- 838 Mohr W, Lehnen N, Ahmerkamp S, Marchant HK, Graf JS, Tschitschko B, Yilmaz P, Littmann
 839 S, Gruber-Vodicka H, Leisch N et al. (2021) Terrestrial-type nitrogen-fixing symbiosis
 840 between seagrass and a marine bacterium. *Nature* 600, 105–109.
 841 <https://doi.org/10.1038/s41586-021-04063-4>.
- 842 Moore JC, Berlow EL, Coleman DC, Ruiter PC, Dong Q, Hastings A, Johnson NC, McCann KS,
 843 Melville K, Morin PJ et al. (2004) Detritus, trophic dynamics and biodiversity. *Ecology*
 844 *Letters* 7, 584–600. <https://doi.org/10.1111/j.1461-0248.2004.00606.x>.
- 845 Müller K (2025) here: a simpler way to find your files. <https://here.r-lib.org>.

- 846 Neely RK, Wetzel RG (1997) Autumnal production by bacteria and autotrophs attached to *Typha*
 847 *latifolia* L. detritus. *Journal of Freshwater Ecology* 12, 253–267.
 848 <https://doi.org/10.1080/02705060.1997.9663533>.
- 849 Norton TA (1992) Dispersal by macroalgae. *British Phycological Journal* 27, 293–301.
 850 <https://doi.org/10.1080/00071619200650271>.
- 851 Norton TA, Mathieson AC (1983) The biology of unattached seaweeds. Round FE, Chapman DJ
 852 (eds.) *Progress in Phycological Research, Vol. 2*, 333–386.
- 853 Ochieng CA, Erftemeijer PLA (1999) Accumulation of seagrass beach cast along the Kenyan
 854 coast: a quantitative assessment. *Aquatic Botany* 65, 221–238.
 855 [https://doi.org/10.1016/s0304-3770\(99\)00042-x](https://doi.org/10.1016/s0304-3770(99)00042-x).
- 856 Olsen JL, Rouzé P, Verhelst B, Lin Y-C, Bayer T, Collen J, Dattolo E, De Paoli E, Dittami S,
 857 Maumus F et al. (2016) The genome of the seagrass *Zostera marina* reveals angiosperm
 858 adaptation to the sea. *Nature* 530, 331–335. <https://doi.org/10.1038/nature16548>.
- 859 Olson JS (1963) Energy storage and the balance of producers and decomposers in ecological
 860 systems. *Ecology* 44, 322–331. <https://doi.org/10.2307/1932179>.
- 861 Patriquin D, Knowles R (1972) Nitrogen fixation in the rhizosphere of marine angiosperms.
 862 *Marine Biology* 16, 49–58. <https://doi.org/10.1007/bf00347847>.
- 863 Pedersen MF, Filbee-Dexter K, Frisk NL, Sárossy Z, Wernberg T (2021) Carbon sequestration
 864 potential increased by incomplete anaerobic decomposition of kelp detritus. *Marine*
 865 *Ecology Progress Series* 660, 53–67. <https://doi.org/10.3354/meps13613>.
- 866 Pedersen MF, Paling EI, Walker DI (1997) Nitrogen uptake and allocation in the seagrass
 867 *Amphibolis antarctica*. *Aquatic Botany* 56, 105–117. [https://doi.org/10.1016/s0304-](https://doi.org/10.1016/s0304-3770(96)01100-x)
 868 [3770\(96\)01100-x](https://doi.org/10.1016/s0304-3770(96)01100-x).
- 869 Pedersen MF, Borum J (1992) Nitrogen dynamics of eelgrass *Zostera marina* during low nutrient
 870 availability. *Marine Ecology Progress Series* 80, 65–73.
 871 <https://doi.org/10.3354/meps080065>.
- 872 Pedersen TL (2025) patchwork: the composer of plots. <https://patchwork.data-imaginist.com>.
- 873 Pellikaan GC (1982) Decomposition processes of eelgrass, *Zostera marina* L. *Hydrobiological*
 874 *Bulletin* 16, 83–92. <https://doi.org/10.1007/bf02255416>.

- 875 Powles JE, Buckley TN, Nicotra AB, Farquhar GD (2006) Dynamics of stomatal water relations
 876 following leaf excision. *Plant, Cell & Environment* 29, 981–992.
 877 <https://doi.org/10.1111/j.1365-3040.2005.01491.x>.
- 878 POWO (2025) Plants of the World Online. Facilitated by the Royal Botanic Gardens, Kew.
 879 <https://powo.science.kew.org>.
- 880 R Core Team (2026) R: a language and environment for statistical computing. R Foundation for
 881 Statistical Computing. <https://www.R-project.org>.
- 882 Rice DL, Tenore KR (1981) Dynamics of carbon and nitrogen during the decomposition of
 883 detritus derived from estuarine macrophytes. *Estuarine, Coastal and Shelf Science* 13, 681–
 884 690. [https://doi.org/10.1016/s0302-3524\(81\)80049-7](https://doi.org/10.1016/s0302-3524(81)80049-7).
- 885 Rodriguez GE, Reed DC, Holbrook SJ (2016) Blade life span, structural investment, and nutrient
 886 allocation in giant kelp. *Oecologia* 182, 397–404. [https://doi.org/10.1007/s00442-016-](https://doi.org/10.1007/s00442-016-3674-6)
 887 3674-6.
- 888 Rohatgi A (2022) WebPlotDigitizer. <https://automeris.io>.
- 889 Rosch KL, Koch MS (2009) Nitrogen and phosphorus recycling by a dominant tropical seagrass
 890 (*Thalassia testudinum*) across a nutrient gradient in Florida Bay. *Bulletin of Marine Science*
 891 84, 1–24.
 892 [https://www.ingentaconnect.com/contentone/umrsmas/bullmar/2009/00000084/00000001/a](https://www.ingentaconnect.com/contentone/umrsmas/bullmar/2009/00000084/00000001/art00001)
 893 [rt00001](https://www.ingentaconnect.com/contentone/umrsmas/bullmar/2009/00000084/00000001/art00001).
- 894 Rothäusler E, Reinwald H, López BA, Tala F, Thiel M (2018) High acclimation potential in
 895 floating *Macrocystis pyrifera* to abiotic conditions even under grazing pressure – a field
 896 study. *Journal of Phycology* 54, 368–379. <https://doi.org/10.1111/jpy.12643>.
- 897 Rothäusler E, Gómez I, Hinojosa IA, Karsten U, Miranda L, Tala F, Thiel M (2011) Kelp rafts in
 898 the Humboldt Current: Interplay of abiotic and biotic factors limit their floating persistence
 899 and dispersal potential. *Limnology and Oceanography* 56, 1751–1763.
 900 <https://doi.org/10.4319/lo.2011.56.5.1751>.
- 901 Rothäusler E, Rugiu L, Tihihonen T, Jormalainen V (2020) It takes two to stay afloat: interplay of
 902 morphology and physiological acclimation ensures long-term floating dispersal of the
 903 bladderwrack *Fucus vesiculosus* (Phaeophyceae, Fucales). *European Journal of Phycology*
 904 55, 242–252. <https://doi.org/10.1080/09670262.2019.1694706>.

- 905 Rublee PA, Roman MR (1982) Decomposition of turtlegrass (*Thalassia testudinum* König) in
 906 flowing sea-water tanks and litterbags: compositional changes and comparison with natural
 907 particulate matter. *Journal of Experimental Marine Biology and Ecology* 58, 47–58.
 908 [https://doi.org/10.1016/0022-0981\(82\)90096-x](https://doi.org/10.1016/0022-0981(82)90096-x).
- 909 Short FT, McRoy CP (1984) Nitrogen uptake by leaves and roots of the seagrass *Zostera marina*
 910 L. *Botanica Marina* 27, 547–555. <https://doi.org/10.1515/botm.1984.27.12.547>.
- 911 Smith BD, Foreman RE (1984) An assessment of seaweed decomposition within a southern
 912 Strait of Georgia seaweed community. *Marine Biology* 84, 197–205.
 913 <https://doi.org/10.1007/bf00393005>.
- 914 Stafford-Bell RE, Chariton AA, Robinson RW (2015) Prolonged buoyancy and viability of
 915 *Zostera muelleri* Irmisch ex Asch. vegetative fragments indicate a strong dispersal
 916 potential. *Journal of Experimental Marine Biology and Ecology* 464, 52–57.
 917 <https://doi.org/10.1016/j.jembe.2014.12.014>.
- 918 Stapel J, Aarts TL, van Duynhoven BHM, de Groot JD, van den Hoogen PHW, Hemminga MA
 919 (1996) Nutrient uptake by leaves and roots of the seagrass *Thalassia hemprichii* in the
 920 Spermonde Archipelago, Indonesia. *Marine Ecology Progress Series* 134, 195–206.
 921 <https://doi.org/10.3354/meps134195>.
- 922 Strother S, Vatta R (1986) The effects of light, cycloheximide and incubation medium on the
 923 senescence of detached seagrass leaves. *Biologia Plantarum* 28, 211–218.
 924 <https://doi.org/10.1007/bf02894599>.
- 925 Tala F, Gómez I, Luna-Jorquera G, Thiel M (2013) Morphological, physiological and
 926 reproductive conditions of rafting bull kelp (*Durvillaea antarctica*) in northern-central
 927 Chile (30°S). *Marine Biology* 160, 1339–1351. <https://doi.org/10.1007/s00227-013-2186-8>.
- 928 Tala F, López BA, Velásquez M, Jeldres R, Macaya EC, Mansilla A, Ojeda J, Thiel M (2019)
 929 Long-term persistence of the floating bull kelp *Durvillaea antarctica* from the South-East
 930 Pacific: potential contribution to local and transoceanic connectivity. *Marine*
 931 *Environmental Research* 149, 67–79. <https://doi.org/10.1016/j.marenvres.2019.05.013>.
- 932 Tala F, Velásquez M, Mansilla A, Macaya EC, Thiel M (2016) Latitudinal and seasonal effects
 933 on short-term acclimation of floating kelp species from the South-East Pacific. *Journal of*
 934 *Experimental Marine Biology and Ecology* 483, 31–41.
 935 <https://doi.org/10.1016/j.jembe.2016.06.003>.

- 936 Thimann KV, Satler SO (1979) Relation between leaf senescence and stomatal closure:
 937 senescence in light. *Proceedings of the National Academy of Sciences* 76, 2295–2298.
 938 <https://doi.org/10.1073/pnas.76.5.2295>.
- 939 Thomas F, Cosse A, Le Panse S, Kloareg B, Potin P, Leblanc C (2014) Kelps feature systemic
 940 defense responses: insights into the evolution of innate immunity in multicellular
 941 eukaryotes. *New Phytologist* 204, 567–576. <https://doi.org/10.1111/nph.12925>.
- 942 Trevathan-Tackett SM, Jeffries TC, Macreadie PI, Manojlovic B, Ralph P (2020) Long-term
 943 decomposition captures key steps in microbial breakdown of seagrass litter. *Science of the*
 944 *Total Environment* 705, 135806. <https://doi.org/10.1016/j.scitotenv.2019.135806>.
- 945 Twigg IM, Baltar F, Hall JR, Hepburn CD (2020) Revealing hydrogen peroxide as an external
 946 stressor in macrophyte-dominated coastal ecosystems. *Oecologia* 193, 583–591.
 947 <https://doi.org/10.1007/s00442-020-04690-0>.
- 948 Vähätalo A, Søndergaard M, Schlüter L, Markager S (1998) Impact of solar radiation on the
 949 decomposition of detrital leaves of eelgrass *Zostera marina*. *Marine Ecology Progress*
 950 *Series* 170, 107–117. <https://doi.org/10.3354/meps170107>.
- 951 van den Brand T (2025) ggh4x: hacks for ‘ggplot2’. <https://github.com/teunbrand/ggh4x>.
- 952 Vandendriessche S, Vincx M, Degraer S (2007) Floating seaweed and the influences of
 953 temperature, grazing and clump size on raft longevity — a microcosm study. *Journal of*
 954 *Experimental Marine Biology and Ecology* 343, 64–73.
 955 <https://doi.org/10.1016/j.jembe.2006.11.010>.
- 956 Viana IG, Saavedra-Hortúa DA, Mtolera M, Teichberg M (2019) Different strategies of nitrogen
 957 acquisition in two tropical seagrasses under nitrogen enrichment. *New Phytologist* 223,
 958 1217–1229. <https://doi.org/10.1111/nph.15885>.
- 959 Waycott M, Biffin E, Les DH (2018) Systematics and evolution of Australian seagrasses in a
 960 global context. Larkum AWD, Kendrick GA, Ralph PJ (eds.) *Seagrasses of Australia:*
 961 *Structure, Ecology and Conservation*, 129–154. [https://doi.org/10.1007/978-3-319-71354-](https://doi.org/10.1007/978-3-319-71354-0_5)
 962 [0_5](https://doi.org/10.1007/978-3-319-71354-0_5).
- 963 Weatherall EJ, Ostrowski A, Lekammudiyanse MU, Hendry RA, Jackson EL (2025) Re-
 964 attachment and viability of vegetative propagules of subtropical intertidal seagrasses.
 965 *Marine Ecology Progress Series* 767, 27–40. <https://doi.org/10.3354/meps14917>.

- 966 Weinberger F, Friedlander M, Hoppe H (1999) Oligoagars elicit a physiological response in
 967 *Gracilaria conferta* (Rhodophyta). *Journal of Phycology* 35, 747–755.
 968 <https://doi.org/10.1046/j.1529-8817.1999.3540747.x>.
- 969 Weinberger F, Friedlander M (2000) Response of *Gracilaria conferta* (Rhodophyta) to
 970 oligoagars results in defense against agar-degrading epiphytes. *Journal of Phycology* 36,
 971 1079–1086. <https://doi.org/10.1046/j.1529-8817.2000.00003.x>.
- 972 White LJ, Norkko A (2025) A path towards appropriate degradation experiments for assessing
 973 carbon sequestration potential of macroalgae. *Journal of Ecology* 113, 2730–2743.
 974 <https://doi.org/10.1111/1365-2745.70107>.
- 975 Wickham H, Averick M, Bryan J, Chang W, McGowan LD, François R, Grolemund G, Hayes
 976 A, Henry L, Hester J (2019) Welcome to the Tidyverse. *Journal of Open Source Software*
 977 4, 1686. <https://doi.org/10.21105/joss.01686>.
- 978 Wilke CO (2021) gggridges: ridgeline plots in ggplot2.
 979 <https://doi.org/10.32614/CRAN.package.gggridges>.
- 980 Wolodzko T (2023) extraDistr: additional univariate and multivariate distributions.
 981 <https://doi.org/10.32614/CRAN.package.extraDistr>.
- 982 Wright LS, Foggo A (2021) Photosynthetic pigments of co-occurring Northeast Atlantic
 983 *Laminaria* spp. are unaffected by decomposition. *Marine Ecology Progress Series* 678,
 984 227–232. <https://doi.org/10.3354/meps13886>.
- 985 Wright LS, Pessarrodona A, Foggo A (2022) Climate-driven shifts in kelp forest composition
 986 reduce carbon sequestration potential. *Global Change Biology* 28, 5514–5531.
 987 <https://doi.org/10.1111/gcb.16299>.
- 988 Wright LS, Kregting L (2023) Genus-specific response of kelp photosynthetic pigments to
 989 decomposition. *Marine Biology* 170, 144. <https://doi.org/10.1007/s00227-023-04289-y>.
- 990 Wright LS, Simpkins T, Filbee-Dexter K, Wernberg T (2024) Temperature sensitivity of detrital
 991 photosynthesis. *Annals of Botany* 133, 17–28. <https://doi.org/10.1093/aob/mcad167>.
- 992 Wright LS (2025) Detrital photosynthesis (Dataset). <https://doi.org/10.5281/zenodo.15054888>.
- 993 Wright LS (2026a) Macroalgal exponential decay (Dataset).
 994 <https://doi.org/10.5281/zenodo.18169931>.
- 995 Wright LS (2026b) A model of macroalgal decomposition. *Ecological Modelling* 519, 111549.
 996 <https://doi.org/10.1016/j.ecolmodel.2026.111549>.

997

Peer Review

